Unmapped Element Reduction Analysis — Wave W003 Phase 1

Project: MINERVA CryoCell / QCELL & RFCELL P&ID — colour-line model
Wave: W003 (Layer Hierarchy), Phase 1
Method: Legend-driven reclassification of previously-unmapped SVG elements.

1. Summary

Metric	Count
Unmapped elements (entering W003, from W002)	**982**
Reclassified into a known category	**870**
Still unresolved	**112**
Reduction achieved	**88.6 %**

The 982 elements that W002 could not bind to a colour-line were re-examined against the extracted legend symbols and against their geometric/shape signature (dot, triangle, bubble, rectangle, leader line, arrow, structural path). 870 of them (88.6 %) were confidently routed to a functional category. The residual **112** are all in a single bucket — `UNRESOLVED_OTHER_COLOUR` — corresponding to the magenta / non-canonical colour family that has no legend entry yet.

2. Reclassified categories (870)

Category	Count	Meaning
`symbol_glyph_or_heatload_marker`	236	Symbol glyphs and heat-load marker fragments
`structure_leader_or_signal_line`	211	Leader lines / signal (dashed) lines
`spec_change_dot_uncoloured`	153	Spec-change dots with no colour fill
`instrument_bubble`	146	Instrument bubbles (circles around tags)
`structure_symbol_path`	57	Structural symbol outlines
`annotation_flow_arrow`	40	Flow-direction arrows used as annotation
`structure_boundary_or_titleblock`	27	Scope boundary / title-block rectangles
Total	**870**	

Representative element IDs per category are stored in `data/model/unmapped_reduction.json → examples`.

3. Still unresolved (112)

Category	Count	Disposition
`UNRESOLVED_OTHER_COLOUR`	112	**DEFERRED** — magenta / non-canonical colour family

These elements (e.g. `path42`, `path44`, `path62`, `path63`) carry a fill colour outside the canonical palette (BLUE→A, CYAN→B, GREEN→W, OLIVE→S, GREY→V, ORANGE→D, RED→E, BLACK→structure). No legend swatch maps to them. They are **honestly retained as unresolved** rather than force-fit, and are tracked in `TODO.md` for a future legend-augmentation pass.

4. Provenance

- Engine: `src/abacus_svg_pid/build_w003_w004.py` (Phase 1)
- Geometry extraction: `src/abacus_svg_pid/geometry.py` (CTM-resolved bbox/centroid/shape)
- Machine-readable output: `data/model/unmapped_reduction.json`